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Androbiome

A Brief Guide to Results Interpretation for the Androbiome and Androbiome Screen Assays

Male Urogenital Microbiome Analysis

For Research and Diagnostic Use

Introduction

This brochure provides an overview of the **Androbiome** and **Androbiome Screen** assays for PCR-based quantification of microbiota composition, interpretation of results, and recommendations for the choice of assays.

About the Assays

The Androbiome assay employs real-time PCR technology to detect and quantify DNA from microorganisms colonizing the male urogenital tract. This method provides comprehensive analysis of the urogenital microbiome, enabling clinicians to make informed diagnostic and treatment decisions.

Key Features

- Quantitative analysis of microbiota composition
- Detection of opportunistic microorganisms
- Identification of sexually transmitted infections (STIs)
- Assessment of sample quality through human DNA quantification
- Evaluation of total bacterial mass (TBM)

Clinical Applications

- Diagnosis of urogenital infections in men
- Assessment of microbiome dysbiosis
- Monitoring treatment efficacy
- Screening for STIs
- Pre-surgical microbiological assessment

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1. Choice of Assay

The choice between **Androbiome** and **Androbiome Screen** depends on the clinical objectives and the level of detail required for diagnosis.

Androbiome (Comprehensive)

The full Androbiome assay provides comprehensive analysis of the male urogenital microbiome. It is recommended for:

- Detailed microbiome profiling
- Diagnosis of complex or recurrent infections
- Pre-treatment baseline assessment
- Monitoring chronic conditions
- Research applications requiring complete microbiome data

Androbiome Screen (Screening)

The Androbiome Screen assay is a streamlined version designed for rapid screening. It is recommended for:

- Initial screening of symptomatic patients
- STI screening programs
- Follow-up testing after treatment
- High-throughput laboratory workflows
- Cost-effective routine testing

Recommendation

When clinical presentation is unclear or when comprehensive microbiome assessment is needed, the full Androbiome assay is preferred. For routine screening and follow-up, Androbiome Screen provides rapid and cost-effective results.

2. Comparison of Assays

The table below shows the microorganisms detected by each assay.

Parameter / Microorganism	Androbiome	Androbiome Screen
Control Parameters		
Human DNA (sample quality control)	✓	✓
Total Bacterial Mass (TBM)	✓	✓
Transit Microorganisms		
Lactobacillus spp.	✓	—
Commensals (Normal Flora)		
Staphylococcus spp.	✓	—
Streptococcus spp.	✓	—
Corynebacterium spp.	✓	—
Opportunistic Microorganisms (BV-associated)		
Gardnerella vaginalis	✓	✓
Atopobium cluster	✓	—
Megasphaera spp. / Veillonella spp. / Dialister spp.	✓	—
Sneathia spp. / Leptotrichia spp. / Fusobacterium spp.	✓	—
Lachnobacterium spp. / Clostridium spp.	✓	—
Mobiluncus spp. / Corynebacterium spp.	✓	—
Peptostreptococcus spp. / Parvimonas spp.	✓	—

Comparison of Assays (Continued)

Parameter / Microorganism	Androbiome	Androbiome Screen
Opportunistic Anaerobes		
Anaerococcus spp.	✓	—
Prevotella bivia / Porphyromonas spp.	✓	—
Eubacterium spp.	✓	—
Haemophilus spp.	✓	—
Pseudomonas aeruginosa / Ralstonia spp. / Burkholderia spp.	✓	—
Enterobacteriaceae spp. / Enterococcus spp.	✓	✓
Mycoplasmas		
Mycoplasma hominis	✓	✓
Ureaplasma urealyticum	✓	✓
Ureaplasma parvum	✓	✓
Candida		
Candida spp.	✓	✓
Sexually Transmitted Infections (STIs)		
Mycoplasma genitalium	✓	✓
Trichomonas vaginalis	✓	✓
Neisseria gonorrhoeae	✓	✓
Chlamydia trachomatis	✓	✓

✓ Included in assay — Not included

3. Biomaterial Sampling

Proper sample collection is essential for accurate results. The following biomaterials can be analyzed:

Acceptable Sample Types

Sample Type	Collection Method	Notes
Urethral scrape	Sterile swab/probe	Primary sample type
First-void urine	Sterile container	10-15 mL, first morning preferred
Ejaculate	Sterile container	After 2-3 days abstinence
Prostatic secretion	After prostate massage	Collected by physician
Post-massage urine	Sterile container	First portion after massage
Biopsy material	Sterile container with saline	For tissue samples

Important

Patient should not urinate for at least 2 hours before urethral sample collection. Antibiotic therapy should be completed at least 2 weeks before sampling for optimal results.

Urethral Scrape Collection

1 **Preparation**
Patient should not urinate for at least 2 hours before collection. Clean the external urethral opening with sterile saline.

2 **Sample Collection**
Insert the sterile probe 2-4 cm into the urethra. Rotate the probe gently while slowly withdrawing to collect epithelial cells.

3 **Transfer to Transport Medium**
Immediately place the probe into the transport tube containing appropriate medium. Break off the handle at the marked line.

4 **Labeling and Storage**
Label the tube with patient information and collection date. Store at 2-8°C and transport to the laboratory within 24 hours.

Urine Collection

1 **Timing**
First morning urine is preferred. Patient should not have urinated for at least 4 hours before collection.

2 **Collection**
Collect the first 10-15 mL of urine stream (first-void) into a sterile container. This portion contains the highest concentration of urethral microorganisms.

3 **Storage**
Close the container tightly. Store at 2-8°C and deliver to laboratory within 6 hours for optimal results.

Ejaculate Collection

1 **Preparation**
Sexual abstinence for 2-3 days before collection is recommended. Patient should urinate and wash hands before collection.

2 **Collection**
Collect the entire ejaculate into a sterile, wide-mouth container by masturbation. Ensure no lubricants or saliva contact the sample.

3 **Transport**
Keep sample at room temperature (20-37°C) and deliver to laboratory within 1 hour. Do not refrigerate semen samples.

Prostatic Secretion Collection

1 **Patient Preparation**
Patient should empty bladder before the procedure. Position patient appropriately for digital rectal examination.

2 **Prostate Massage**
Physician performs digital prostate massage to express prostatic secretion through the urethral opening.

3 **Collection**
Collect the expressed secretion directly into a sterile container or onto a sterile swab. For post-massage urine, collect immediately after massage.

Note

If insufficient prostatic secretion is obtained, collect the first portion of urine immediately after massage (post-massage urine) as an alternative sample.

Sample Quality Requirements

Sample Type	Human DNA Requirement	Storage Temperature	Max. Transport Time
Urethral scrape	≥ 10^4 GE/sample	2-8°C	24 hours
First-void urine	≥ 10^4 GE/sample	2-8°C	6 hours
Ejaculate	≥ 10^4 GE/sample	20-37°C	1 hour
Prostatic secretion	≥ 10^3 GE/sample	2-8°C	24 hours
Post-massage urine	≥ 10^4 GE/sample	2-8°C	6 hours

GE = Genome Equivalents

Insufficient Sample Quality

If the Human DNA value is below the threshold, the sample is considered inadequate for analysis. This may indicate:

- Improper sample collection technique
- Insufficient cellular material
- Sample degradation during transport

A new sample should be collected following proper procedures.

Optimal Results

For best results, ensure samples are collected following all guidelines, properly labeled, and transported under appropriate conditions within the specified timeframes.

4. Analysis of Results

Results are presented as quantitative values expressed in genome equivalents per sample (GE/sample) or as logarithmic values (lg GE/sample). The interpretation considers both individual values and the overall microbiome composition.

Result Categories

Normal / Not detected Moderate elevation Abnormal / Detected (for STIs)

Control Parameters

Parameter	Purpose	Interpretation
Human DNA	Sample quality control	Must meet minimum threshold for valid results
Total Bacterial Mass (TBM)	Overall bacterial load	Baseline for calculating relative abundances

Interpretation Principles

- **Absolute values:** Expressed as GE/sample or lg GE/sample
- **Relative abundance:** Percentage of TBM for each microorganism group
- **Reference ranges:** Based on population studies of healthy males
- **Clinical correlation:** Results should be interpreted alongside clinical presentation

Microorganism Groups Interpretation

Group	Clinical Significance	Action
Lactobacillus spp.	Transit flora, typically from partner; generally not pathogenic in males	Usually no treatment required
Commensals (Staphylococcus, Streptococcus, Corynebacterium)	Normal urethral flora; elevated levels may indicate dysbiosis	Monitor if elevated; treat if symptomatic
BV-associated (Gardnerella, Atopobium, etc.)	Associated with bacterial vaginosis in partners; may cause urethritis	Consider treatment if symptomatic or partner has BV
Opportunistic anaerobes	May cause infection when host defenses compromised	Treat based on clinical presentation
Mycoplasmas (M. hominis, Ureaplasma)	May cause NGU; clinical significance depends on load	Treat if $>10^4$ GE/sample and symptomatic
Candida spp.	Fungal infection; may cause balanitis	Antifungal treatment if detected and symptomatic
STIs (M. genitalium, T. vaginalis, N. gonorrhoeae, C. trachomatis)	Sexually transmitted pathogens; always clinically significant	**Always treat if detected**; partner notification required

STI Detection

Any detection of **Mycoplasma genitalium**, **Trichomonas vaginalis**, **Neisseria gonorrhoeae**, or **Chlamydia trachomatis** requires immediate treatment and partner notification, regardless of symptoms.

Reference Ranges

The following reference ranges apply to urethral scrape samples. Ranges may vary for other sample types.

Parameter	Normal	Moderate	Abnormal
Human DNA	$\geq 10^4$ GE	10^3 - 10^4 GE	$< 10^3$ GE
Total Bacterial Mass	10^3 - 10^6 GE	10^6 - 10^7 GE	$> 10^7$ GE
Commensals (% of TBM)	< 50%	50-80%	> 80%
Opportunistic bacteria	$< 10^4$ GE	10^4 - 10^5 GE	$> 10^5$ GE
Mycoplasmas	$< 10^3$ GE	10^3 - 10^4 GE	$> 10^4$ GE
Candida spp.	Not detected	10^2 - 10^4 GE	$> 10^4$ GE
STIs	Not detected		**Detected**

Clinical Context

Reference ranges are guidelines. Clinical interpretation should always consider:

- Patient symptoms and clinical presentation
- Sample type and collection quality
- Previous test results and treatment history
- Partner's health status

Clinical Recommendations

Normal Results

- No pathogenic microorganisms detected
- Normal commensal flora composition
- No STIs detected
- **Action:** No treatment required; routine follow-up as indicated

Dysbiosis / Moderate Abnormalities

- Elevated opportunistic bacteria
- Imbalanced microbiome composition
- Moderate mycoplasma levels
- **Action:** Consider treatment if symptomatic; repeat testing in 4-6 weeks if asymptomatic

STI Detection / Significant Pathology

- Any STI pathogen detected
- High levels of pathogenic bacteria
- Significant fungal infection
- **Action:** Immediate treatment according to guidelines; partner notification and testing; follow-up testing after treatment

Follow-up Testing

Recommended follow-up intervals:

- **After STI treatment:** 3-4 weeks post-treatment
- **After bacterial infection treatment:** 4-6 weeks
- **Chronic conditions:** Every 3-6 months as clinically indicated
- **Asymptomatic dysbiosis:** 6-12 weeks to monitor resolution

5. Example Result Forms

The following pages present example result forms demonstrating various clinical scenarios. Each example includes:

- Complete microbiome analysis results
- Color-coded status indicators
- Quantitative values (GE/sample and lg values)
- Clinical interpretation notes

How to Read the Result Forms

Normal / Not detected   Moderate / Attention   Abnormal / Detected

Column	Description
Microorganism	Name of the detected microorganism or group
Result (GE)	Absolute value in genome equivalents per sample
lg (GE)	Logarithmic value (base 10)
Reference	Expected normal range
Status	Color-coded interpretation

Note

All example results are fictional and for educational purposes only. Actual patient results should be interpreted by qualified healthcare professionals.

Example 1: Normal Microbiome (Urethral Scrape)

Patient ID:	AB-2024-001	Sample Type:	Urethral scrape
Collection Date:	15.01.2024	Analysis Date:	16.01.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	2.5×10^5	5.4	Adequate
Total Bacterial Mass (TBM)	1.8×10^4	4.3	Normal
Transit Microorganisms			
Lactobacillus spp.	Not detected	-	Normal
Commensals			
Staphylococcus spp.	5.2×10^3	3.7	Normal
Streptococcus spp.	2.1×10^3	3.3	Normal
Corynebacterium spp.	3.8×10^3	3.6	Normal
Opportunistic Microorganisms			
Gardnerella vaginalis	Not detected	-	Normal
Enterobacteriaceae / Enterococcus	Not detected	-	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Normal urogenital microbiome. No pathogenic microorganisms detected. No treatment required.

Example 2: Chlamydia trachomatis Detected

Patient ID:	AB-2024-002	Sample Type:	Urethral scrape
Collection Date:	18.01.2024	Analysis Date:	19.01.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	3.1×10^5	5.5	Adequate
Total Bacterial Mass (TBM)	4.2×10^5	5.6	Elevated
Transit Microorganisms			
Lactobacillus spp.	Not detected	-	Normal
Commensals			
Staphylococcus spp.	8.4×10^3	3.9	Normal
Streptococcus spp.	4.2×10^3	3.6	Normal
Corynebacterium spp.	6.1×10^3	3.8	Normal
Opportunistic Microorganisms			
Gardnerella vaginalis	Not detected	-	Normal
Enterobacteriaceae / Enterococcus	2.3×10^3	3.4	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	1.8×10^5	5.3	DETECTED

Interpretation

Chlamydia trachomatis detected. Immediate antibiotic treatment required. Partner notification and testing mandatory. Follow-up testing recommended 3-4 weeks after treatment completion.

Example 3: Neisseria gonorrhoeae Detected

Patient ID:	AB-2024-003	Sample Type:	Urethral scrape
Collection Date:	20.01.2024	Analysis Date:	21.01.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	4.8×10^{5}	5.7	Adequate
Total Bacterial Mass (TBM)	8.5×10^{6}	6.9	High
Transit Microorganisms			
Lactobacillus spp.	Not detected	-	Normal
Commensals			
Staphylococcus spp.	1.2×10^{4}	4.1	Normal
Streptococcus spp.	5.8×10^{3}	3.8	Normal
Corynebacterium spp.	7.3×10^{3}	3.9	Normal
Opportunistic Microorganisms			
Gardnerella vaginalis	Not detected	-	Normal
Enterobacteriaceae / Enterococcus	4.1×10^{3}	3.6	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	7.2×10^{6}	6.9	DETECTED
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Neisseria gonorrhoeae detected. Urgent antibiotic treatment required per local guidelines. Partner notification mandatory. Test of cure recommended 2 weeks post-treatment.

Example 4: Elevated Ureaplasma urealyticum

Patient ID:	AB-2024-004	Sample Type:	First-void urine
Collection Date:	22.01.2024	Analysis Date:	23.01.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	1.9×10^5	5.3	Adequate
Total Bacterial Mass (TBM)	3.6×10^5	5.6	Elevated
Transit Microorganisms			
Lactobacillus spp.	Not detected	-	Normal
Commensals			
Staphylococcus spp.	6.7×10^3	3.8	Normal
Streptococcus spp.	3.4×10^3	3.5	Normal
Corynebacterium spp.	5.2×10^3	3.7	Normal
Opportunistic Microorganisms			
Gardnerella vaginalis	Not detected	-	Normal
Enterobacteriaceae / Enterococcus	Not detected	-	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	2.8×10^5	5.4	Elevated
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Ureaplasma urealyticum elevated (>10^4 GE). If patient is symptomatic (dysuria, discharge), treatment is recommended. If asymptomatic, consider partner status and clinical context.

Example 5: Mycoplasma genitalium Detected

Patient ID:	AB-2024-005	Sample Type:	Urethral scrape
Collection Date:	25.01.2024	Analysis Date:	26.01.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	2.2×10^5	5.3	Adequate
Total Bacterial Mass (TBM)	2.1×10^5	5.3	Elevated
Transit Microorganisms			
Lactobacillus spp.	Not detected	-	Normal
Commensals			
Staphylococcus spp.	4.5×10^3	3.7	Normal
Streptococcus spp.	2.8×10^3	3.4	Normal
Corynebacterium spp.	3.9×10^3	3.6	Normal
Opportunistic Microorganisms			
Gardnerella vaginalis	Not detected	-	Normal
Enterobacteriaceae / Enterococcus	Not detected	-	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	Not detected	-	Normal
STIs			
Mycoplasma genitalium	8.4×10^4	4.9	DETECTED
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Mycoplasma genitalium detected. Treatment required. Consider macrolide resistance testing if available. Partner notification and testing recommended. Follow-up testing 3-4 weeks post-treatment.

Example 6: Trichomonas vaginalis Detected

Patient ID:	AB-2024-006	Sample Type:	First-void urine
Collection Date:	28.01.2024	Analysis Date:	29.01.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	3.5×10^5	5.5	Adequate
Total Bacterial Mass (TBM)	5.2×10^5	5.7	Elevated
Commensals			
Staphylococcus spp.	7.8×10^3	3.9	Normal
Streptococcus spp.	4.1×10^3	3.6	Normal
Corynebacterium spp.	5.6×10^3	3.7	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	4.3×10^5	5.6	DETECTED
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Trichomonas vaginalis detected. Antiprotozoal treatment required. Partner notification and concurrent treatment mandatory. Retest recommended 3-4 weeks post-treatment.

Example 7: Multiple STI Co-infection

Patient ID:	AB-2024-007	Sample Type:	Urethral scrape
Collection Date:	01.02.2024	Analysis Date:	02.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	4.1×10^5	5.6	Adequate
Total Bacterial Mass (TBM)	1.2×10^7	7.1	High
Commensals			
Staphylococcus spp.	1.5×10^4	4.2	Elevated
Streptococcus spp.	8.2×10^3	3.9	Normal
Corynebacterium spp.	9.4×10^3	4.0	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	5.8×10^6	6.8	DETECTED
Chlamydia trachomatis	2.4×10^5	5.4	DETECTED

Interpretation

Multiple STI co-infection: N. gonorrhoeae and C. trachomatis. Urgent dual antibiotic therapy required. Partner notification mandatory. Consider testing for other STIs (HIV, syphilis). Follow-up testing for both pathogens.

Example 8: Candida Infection

Patient ID:	AB-2024-008	Sample Type:	Urethral scrape
Collection Date:	05.02.2024	Analysis Date:	06.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	2.8×10^5	5.4	Adequate
Total Bacterial Mass (TBM)	2.4×10^4	4.4	Normal
Commensals			
Staphylococcus spp.	6.2×10^3	3.8	Normal
Streptococcus spp.	3.1×10^3	3.5	Normal
Corynebacterium spp.	4.8×10^3	3.7	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	3.6×10^5	5.6	Detected
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Candida spp. detected. If symptomatic (balanitis, itching, discharge), antifungal treatment recommended. Consider partner evaluation if recurrent infections.

Example 9: Gardnerella vaginalis Elevated

Patient ID:	AB-2024-009	Sample Type:	Ejaculate
Collection Date:	08.02.2024	Analysis Date:	09.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	1.8×10^6	6.3	Adequate
Total Bacterial Mass (TBM)	4.5×10^6	6.7	Elevated
Commensals			
Staphylococcus spp.	8.5×10^3	3.9	Normal
Streptococcus spp.	5.2×10^3	3.7	Normal
Corynebacterium spp.	6.8×10^3	3.8	Normal
Opportunistic Microorganisms			
Gardnerella vaginalis	3.8×10^6	6.6	Elevated
Enterobacteriaceae / Enterococcus	2.1×10^3	3.3	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Gardnerella vaginalis elevated. Partner likely has bacterial vaginosis. Consider treatment if patient symptomatic. Partner evaluation and treatment strongly recommended.

Example 10: Insufficient Sample Quality

Patient ID:	AB-2024-010	Sample Type:	Urethral scrape
Collection Date:	10.02.2024	Analysis Date:	11.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	5.2×10^2	2.7	INSUFFICIENT
Total Bacterial Mass (TBM)	1.1×10^3	3.0	Low
Commensals			
Staphylococcus spp.	–	–	N/A
Streptococcus spp.	–	–	N/A
Corynebacterium spp.	–	–	N/A
Mycoplasmas			
Mycoplasma hominis	–	–	N/A
Ureaplasma urealyticum	–	–	N/A
Ureaplasma parvum	–	–	N/A
STIs			
Mycoplasma genitalium	–	–	N/A
Trichomonas vaginalis	–	–	N/A
Neisseria gonorrhoeae	–	–	N/A
Chlamydia trachomatis	–	–	N/A

Interpretation

Sample quality insufficient for analysis. Human DNA below minimum threshold ($<10^4$ GE). Results cannot be interpreted. New sample collection required following proper technique.

Example 11: Prostatic Secretion - Chronic Prostatitis

Patient ID:	AB-2024-011	Sample Type:	Prostatic secretion
Collection Date:	12.02.2024	Analysis Date:	13.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	8.5×10^4	4.9	Adequate
Total Bacterial Mass (TBM)	2.8×10^6	6.4	Elevated
Commensals			
Staphylococcus spp.	3.5×10^4	4.5	Elevated
Streptococcus spp.	1.8×10^4	4.3	Elevated
Corynebacterium spp.	2.1×10^4	4.3	Elevated
Opportunistic Microorganisms			
Enterobacteriaceae / Enterococcus	1.5×10^5	5.2	Elevated
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	4.2×10^4	4.6	Moderate
Ureaplasma parvum	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Pattern consistent with chronic bacterial prostatitis. Elevated Enterobacteriaceae and commensals. Consider targeted antibiotic therapy based on predominant flora.

Example 12: Microbiome Dysbiosis

Patient ID:	AB-2024-012	Sample Type:	First-void urine
Collection Date:	15.02.2024	Analysis Date:	16.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	2.1×10^5	5.3	Adequate
Total Bacterial Mass (TBM)	8.2×10^6	6.9	High
Commensals			
Staphylococcus spp.	5.8×10^5	5.8	High
Streptococcus spp.	2.4×10^5	5.4	High
Corynebacterium spp.	1.8×10^5	5.3	High
Opportunistic Microorganisms			
Gardnerella vaginalis	4.5×10^4	4.7	Moderate
Enterobacteriaceae / Enterococcus	2.8×10^4	4.4	Moderate
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	7.5×10^3	3.9	Moderate
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Significant microbiome dysbiosis. Elevated bacterial mass with overgrowth of commensals. No STIs detected. If symptomatic, consider probiotic support and lifestyle modifications. Repeat testing in 6-8 weeks.

Example 13: Mycoplasma hominis Elevated

Patient ID:	AB-2024-013	Sample Type:	Urethral scrape
Collection Date:	18.02.2024	Analysis Date:	19.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	2.4×10^5	5.4	Adequate
Total Bacterial Mass (TBM)	3.1×10^5	5.5	Elevated
Mycoplasmas			
Mycoplasma hominis	1.8×10^5	5.3	Elevated
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Mycoplasma hominis elevated (>10^4 GE). Treatment recommended if symptomatic. Partner evaluation advised.

Example 14: Post-Treatment Follow-up (Cleared)

Patient ID:	AB-2024-014	Sample Type:	Urethral scrape
Collection Date:	20.02.2024	Analysis Date:	21.02.2024
Previous Dx:	C. trachomatis (25.01.2024)	Treatment:	Azithromycin completed

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	3.2×10^5	5.5	Adequate
Total Bacterial Mass (TBM)	2.1×10^4	4.3	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Treatment successful. Previous C. trachomatis infection has been cleared. Normal microbiome restored. No further treatment required.

Example 15: Androbiome Screen - Normal

Patient ID:	ABS-2024-015	Sample Type:	First-void urine
Test:	Androbiome Screen	Analysis Date:	22.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	2.8×10^{5}	5.4	Adequate
Total Bacterial Mass (TBM)	1.5×10^{4}	4.2	Normal
Opportunistic Microorganisms			
Gardnerella vaginalis	Not detected	–	Normal
Enterobacteriaceae / Enterococcus	Not detected	–	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	–	Normal
Ureaplasma urealyticum	Not detected	–	Normal
Ureaplasma parvum	Not detected	–	Normal
Candida			
Candida spp.	Not detected	–	Normal
STIs			
Mycoplasma genitalium	Not detected	–	Not detected
Trichomonas vaginalis	Not detected	–	Not detected
Neisseria gonorrhoeae	Not detected	–	Not detected
Chlamydia trachomatis	Not detected	–	Not detected

Interpretation

Normal screening result. No STIs or pathogenic microorganisms detected. No treatment required.

Example 16: Androbiome Screen - Chlamydia Detected

Patient ID:	ABS-2024-016	Sample Type:	First-void urine
Test:	Androbiome Screen	Analysis Date:	25.02.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	3.5×10^5	5.5	Adequate
Total Bacterial Mass (TBM)	2.8×10^5	5.4	Elevated
Opportunistic Microorganisms			
Gardnerella vaginalis	Not detected	-	Normal
Enterobacteriaceae / Enterococcus	Not detected	-	Normal
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	1.2×10^5	5.1	DETECTED

Interpretation

Chlamydia trachomatis detected. Immediate treatment required. Partner notification mandatory.

Example 17: Transit Flora (Lactobacillus)

Patient ID:	AB-2024-017	Sample Type:	Urethral scrape
Collection Date:	28.02.2024	Analysis Date:	01.03.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	2.6×10^5	5.4	Adequate
Total Bacterial Mass (TBM)	5.8×10^5	5.8	Elevated
Transit Microorganisms			
Lactobacillus spp.	4.2×10^5	5.6	Present
Commensals			
Staphylococcus spp.	5.4×10^3	3.7	Normal
Streptococcus spp.	2.8×10^3	3.4	Normal
Corynebacterium spp.	4.1×10^3	3.6	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Lactobacillus detected - transit flora from partner. Not pathogenic in males. No treatment required. Normal finding after recent sexual contact.

Example 18: Ureaplasma parvum Elevated

Patient ID:	AB-2024-018	Sample Type:	Ejaculate
Collection Date:	05.03.2024	Analysis Date:	06.03.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	1.5×10^6	6.2	Adequate
Total Bacterial Mass (TBM)	4.8×10^5	5.7	Elevated
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	3.5×10^5	5.5	Elevated
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Ureaplasma parvum elevated. Clinical significance lower than U. urealyticum. Treat if symptomatic or investigating infertility. Partner evaluation may be beneficial.

Example 19: Complex Polymicrobial Infection

Patient ID:	AB-2024-019	Sample Type:	Urethral scrape
Collection Date:	08.03.2024	Analysis Date:	09.03.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	3.8×10^{5}	5.6	Adequate
Total Bacterial Mass (TBM)	2.5×10^{7}	7.4	Very High
Opportunistic Microorganisms			
Gardnerella vaginalis	8.5×10^{5}	5.9	Elevated
Enterobacteriaceae / Enterococcus	2.8×10^{5}	5.4	Elevated
Mycoplasmas			
Mycoplasma hominis	4.5×10^{4}	4.7	Moderate
Ureaplasma urealyticum	1.2×10^{5}	5.1	Elevated
Ureaplasma parvum	Not detected	-	Normal
Candida			
Candida spp.	5.2×10^{3}	3.7	Low positive
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Complex polymicrobial infection. Multiple elevated opportunists and mycoplasmas. Consider broad-spectrum antibiotic therapy. Partner evaluation essential. Follow-up testing required.

Example 20: Biopsy Sample Analysis

Patient ID:	AB-2024-020	Sample Type:	Prostate biopsy
Collection Date:	10.03.2024	Analysis Date:	11.03.2024

Microorganism	Result (GE)	lg (GE)	Status
Control Parameters			
Human DNA	5.2×10^5	5.7	Adequate
Total Bacterial Mass (TBM)	8.4×10^4	4.9	Normal
Commensals			
Staphylococcus spp.	2.1×10^4	4.3	Moderate
Streptococcus spp.	8.5×10^3	3.9	Normal
Corynebacterium spp.	1.2×10^4	4.1	Normal
Opportunistic Microorganisms			
Enterobacteriaceae / Enterococcus	3.8×10^4	4.6	Moderate
Mycoplasmas			
Mycoplasma hominis	Not detected	-	Normal
Ureaplasma urealyticum	Not detected	-	Normal
Ureaplasma parvum	Not detected	-	Normal
STIs			
Mycoplasma genitalium	Not detected	-	Not detected
Trichomonas vaginalis	Not detected	-	Not detected
Neisseria gonorrhoeae	Not detected	-	Not detected
Chlamydia trachomatis	Not detected	-	Not detected

Interpretation

Tissue sample analysis. Low-grade bacterial presence consistent with chronic prostatitis. No STIs detected. Correlate with histopathology findings.

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Disclaimer

This brochure is intended for healthcare professionals. Results should be interpreted in conjunction with clinical findings and patient history. Treatment decisions should follow current local and international guidelines.

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